Short- and Long-Horizon Behavioral Factors

Daniel, Hirshleifer and Sun (2020), *RFS*

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Two biases, two horizons

	Overconfidence	**Limited attention**
Signal	Overconfident about their private signals, and overreact to them	Underreact to public information arriving at high frequency, such as quarterly earnings (Hirshleifer and Teoh 2003; DellaVigna and Pollet 2009)
Correction	Unwilling to correct as public news arrives. Public news can even increase overconfidence (Daniel, Hirshleifer and Subrahmanyam 1998; Gervais and Odean 2001)	Corrected as the next few quarterly earnings are announced (Bernard and Thomas 1990)
Horizon	years	quarters

Why do behavioral biases create factors?

A factor needs comovement. In behavioral models it has two sources:

1. Commonality in stock mispricing (Barberis and Shleifer 2003)
2. Commonality in investor errors when interpreting signals about fundamental economic factors
   (Daniel, Hirshleifer and Subrahmanyam 2001)

Mispricing predicts future returns because it is later corrected.

So, just as firms exposed to systematic risk factors earn a risk premium, firms heavily exposed to behavioral factors earn a conditional return premium (Hirshleifer and Jiang 2010).

FIN: a proxy for long-horizon mispricing

Stein (1996): managers know the intrinsic value of their own firm.

Overpriced	$\rightarrow$ issue equity
Underpriced	$\rightarrow$ repurchase

Investment need not change. This is pure timing.

If investors were rational (Myers-Majluf), financing would not predict returns. It does, and it does so persistently.

Why financing catches only the long horizon

Issuing and repurchasing is slow and costly:

- disclosure, legal and underwriting costs
- fixed costs of starting a programme
- informational barriers

Firms can only time mispricing that persists.

And managers see *total* mispricing, whatever bias caused it, so FIN is a catch-all for long-horizon distortions.

FIN: from data to factor

Universe. CRSP NYSE, AMEX and NASDAQ common stocks (share codes 10 and 11), excluding financials and negative book equity. 1972 to 2014.

Two issuance measures, taken unchanged from earlier papers.

NSI	1 year	log growth in split-adjusted shares outstanding (Pontiff and Woodgate 2008)
CSI	5 years	5-year growth in market equity minus the 5-year stock return, in logs (Daniel and Titman 2006)

Sort, each June. Size at the NYSE median; financing into low, middle, high from an index of the NSI and CSI rankings. Six value-weighted portfolios, held July to June.

$$\text{FIN} = (r_{SL} + r_{BL})/2 \; - \; (r_{SH} + r_{BH})/2$$

PEAD: a proxy for short-horizon mispricing

Prices underreact to earnings news and drift afterwards.

Evidence that this is attention, not risk. Announcements released:

- on Fridays
- on days crowded with other announcements
- outside trading hours

$\rightarrow$ weaker immediate reaction, stronger drift.

The day of the week is random. Risk cannot explain that.

PEAD: from data to factor

Universe. Same stocks, excluding financials. Announcement date is Compustat quarterly item RDQ.

Earnings surprise is the 4-day cumulative abnormal return around the most recent announcement (Chan, Jegadeesh and Lakonishok 1996):

$$\mathrm{CAR}_i = \sum_{d=-2}^{+1} (R_{i,d} - R_{m,d})$$

Screens. At least two valid daily returns in the window; RDQ at least two trading days before month end; drop firms with no announcement in the past six months.

Sort, each month. Size at the NYSE median; surprise into low, middle, high at the NYSE 20th and 80th percentiles. Six value-weighted portfolios, held one month.

$$\mathrm{PEAD} = (r_{SH} + r_{BH})/2 \; - \; (r_{SL} + r_{BL})/2$$

The model

MKT	+	FIN	+	PEAD
rational risk premium		persistent mispricing (years)		transient mispricing (quarters)

Risk, plus two biases. Markets are not assumed irrational.
MKT absorbs whatever premium really is compensation for risk.

Three characteristics: size, financing, earnings surprise.

The competitors

Model	Factors	Characteristics
NM4 (Novy-Marx 2013)	4	5
FF5 (Fama-French 2015)	5	4
HXZ4 (Hou, Xue and Zhang 2015)	4	3
SY4 (Stambaugh and Yuan 2017)	4	12
BF3 (this paper)	**3**	**3**

SY4 has the same architecture: market, one long-horizon mispricing factor, one short-horizon one.

But SY4 averages the ranks of eleven anomalies that already worked.
This paper reasoned first.

Result 1: the factors themselves

	Mean (%/mo)	t	Sharpe
FIN	0.80	4.60	0.20
PEAD	0.65	7.91	0.35
MKT	0.53	2.62	0.12
HML	0.41	3.14	0.14
MGMT	0.67	5.24	0.23

Both clear the $t > 3$ hurdle of Harvey, Liu and Zhu (2016).

1972 to 2014, 510 months.

Result 2: spanning tests

FIN and PEAD alone price 7 of 10 competing factors:
HML, MOM, CMA, RMW, ROE, MGMT, PERF.

The reverse fails. Against *all* other factors at once,
PEAD still earns $\alpha = 0.58\%$ per month, $t = 6.76$.

Maximum Sharpe ratio	
All other factors, no FIN or PEAD	0.54
Adding FIN and PEAD	0.65

Result 3: 34 anomalies

Number of significant α's at 5%. Fewer is better.

	Short (12)	Long (22)	All (34)
BF3	**0**	3	**3**
FF5	11	7	18
NM4	2	3	5
HXZ4	1	5	6
SY4	4	3	7

On the twelve short-horizon anomalies the GRS test cannot reject BF3 ($p = 0.32$); it rejects all four rivals at 1%.

PEAD does this on its own.

Result 4: mispricing, or risk after all?

Recall that the data cannot separate the two stories.
Except here. Short-sale constraints are asymmetric.

Overpricing is harder to correct, so the loadings should be larger on the short leg. Risk predicts no such thing.

Average loading	Long leg	Short leg	Cases
β_{PEAD}, short-horizon	$+0.31$	-0.51	11 of 12
β_{FIN}, long-horizon	$+0.03$	-0.27	15 of 22

Contribution

Paper goes further in three key ways.

1. A strong dichotomy between short- and long-horizon anomalies: short-horizon anomalies predominantly explained by the PEAD-based factor, long-horizon anomalies predominantly explained by the financing factor.

2. The behavioral factors are based on theoretical arguments as to what variables should capture long- and short-horizon mispricing.

3. The factor model provides a better fit to a wide set of anomalies and factors.

Thank you